

Circular #3

Reference Data	
Mass of Earth	$5.98 \times 10^{24} \text{ kg}$
Mass of Sun	$1.99 \times 10^{30} \text{ kg}$
Radius of Earth	$6.38 \times 10^6 \text{ m}$

1. When a spaceship is orbiting the earth and wants to land, it fires *retro thrusters*, which are rockets that fire in the opposite direction than the spaceship is moving. This causes the spaceship to slow down. Explain why slowing down will cause the ship to come in for a landing.
2. If the Sun and its gravity were to suddenly disappear, what would happen to the motion of the earth?
3. A certain astronaut has a mass of 92 kg.
 - (a) Find the force of gravity acting on the astronaut when she is on the earth.
 - (b) The International Space Station (ISS) orbits at a radius of $6.78 \times 10^6 \text{ m}$ from the center of the Earth. Calculate the force of gravity acting on the astronaut when she is aboard the ISS.
 - (c) It should be clear from these calculations that even though the astronaut has a moderately lower force of gravity, it is nowhere near zero gravity. Why, then, does the astronaut experience “weightlessness”?

Name: _____

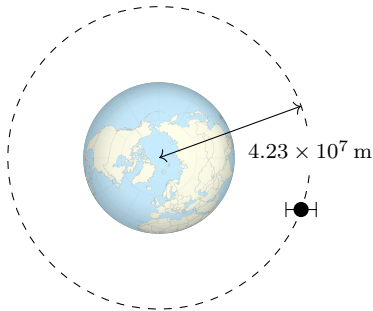
Date: _____

Period: _____

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4. A 2500-kg weather satellite orbits about 22,000 miles above the Earth's equator. This is a radius of $4.23 \times 10^7 \text{ m}$ from the center of the Earth.

(a) Find the force of gravity between the earth and the satellite.



(b) Find the tangential velocity of the satellite. (*use your answer from part a*)

(c) Find how long it takes for the satellite to complete one rotation (*use your answer from part b*)

(d) Your answer to part c is in seconds. Convert it to hours.

(e) Look at your answer to part d. This type of orbit is known as a “*geosynchronous orbit*”. Can you explain why?